



BIG DATA

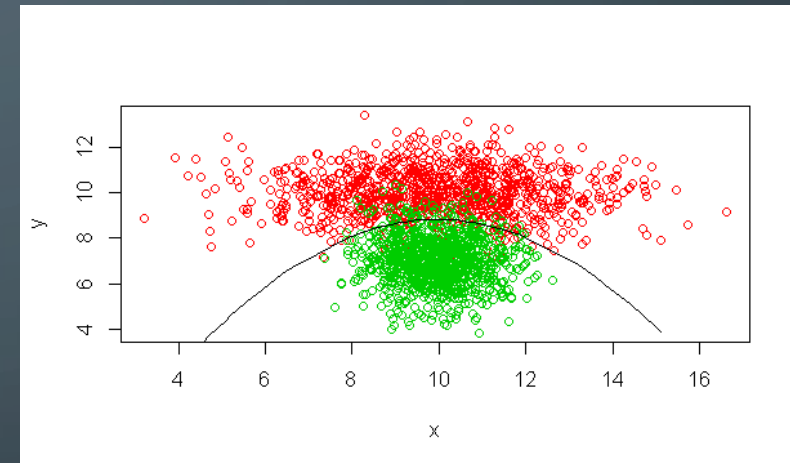
CLASSIFIERS

LEARNING OBJECTIVES

- At the end of this presentation, you should be able to explain
 - how a Naive Bayes classifier works;
 - how SVM works;
 - how decision trees and random forests work;
 - roughly how they compare in terms of accuracy;
 - roughly how they compare in terms of computational complexity.

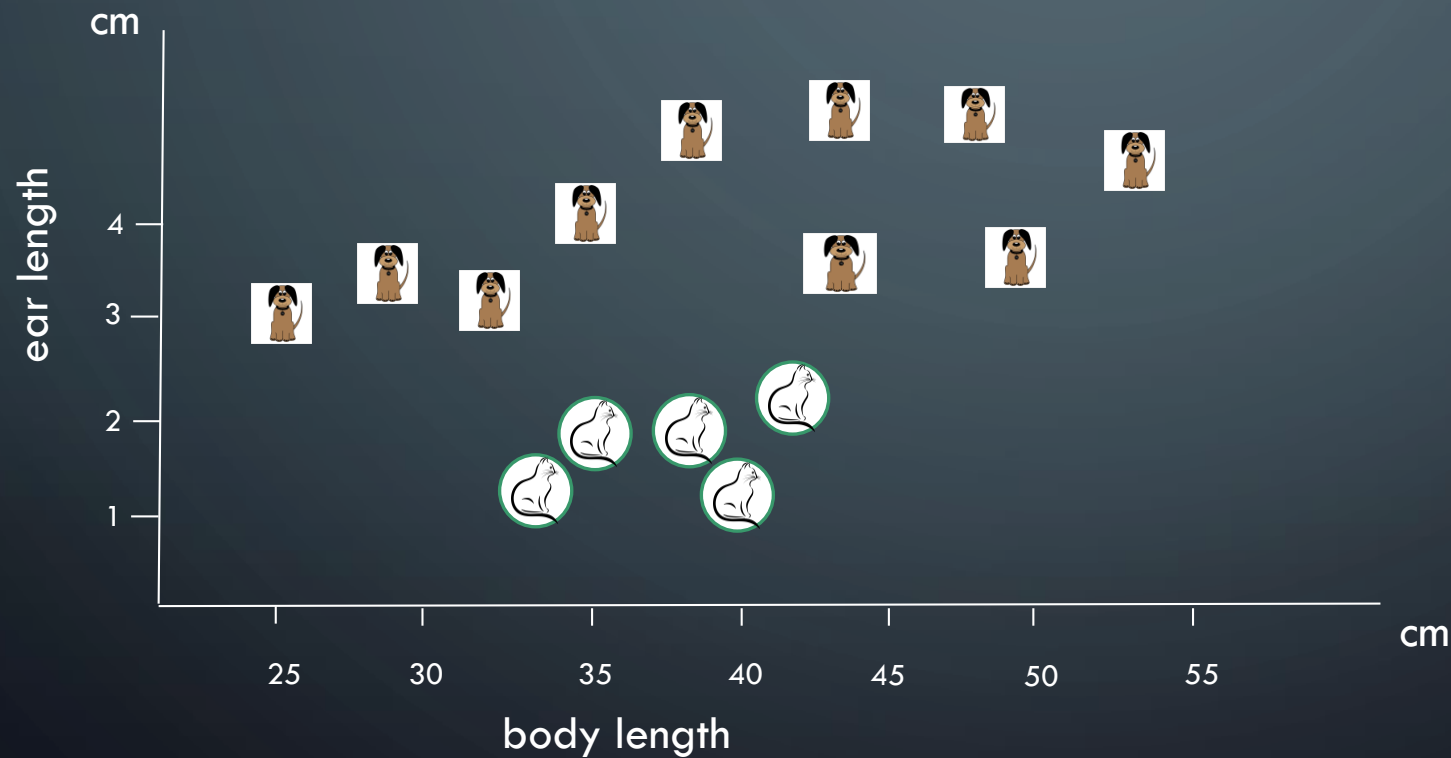
CLASSIFIERS

- Naive Bayes
- Support Vector Machines (SVM)
- Decision Trees
- Random Forest (RF)
-(e.g. deep learning)



unsupervised: **k-means** clustering

BUT FIRST: SCATTERPLOTS

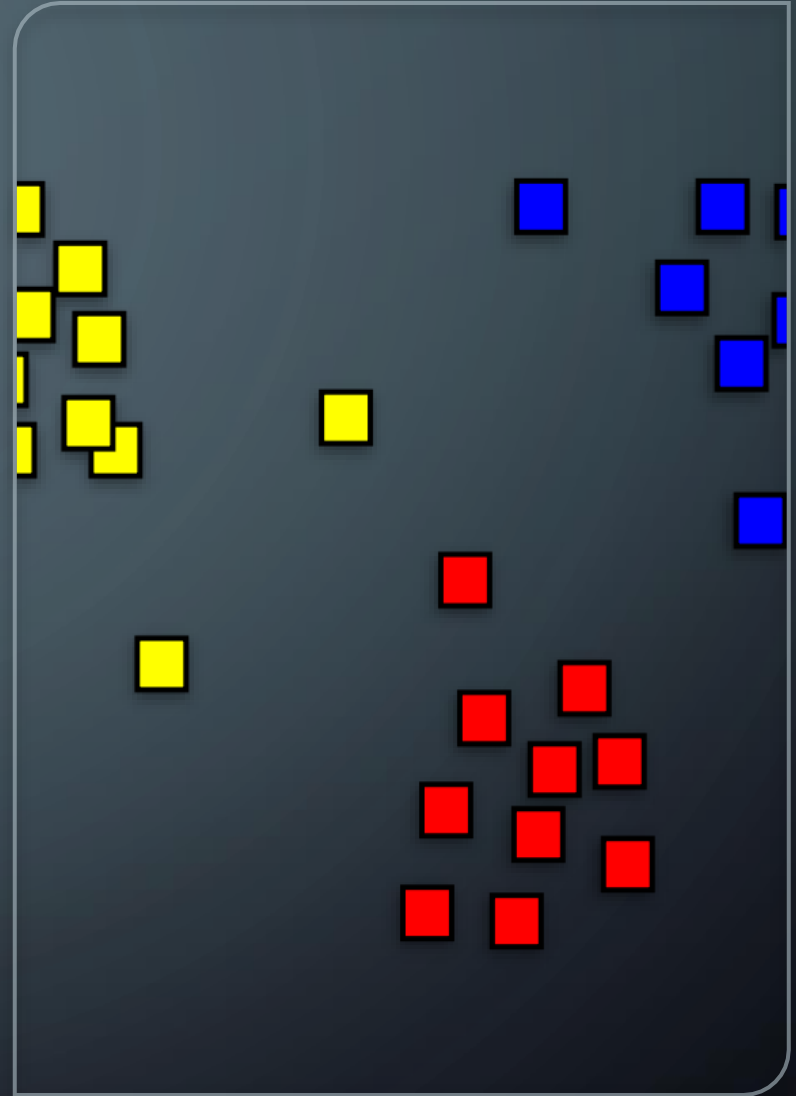


attribute
or feature

type	length	ears	ears
dog	60cm	pointy	5cm
cat	45cm	pointy	2cm
dog	35cm	floppy	1cm

NAÏVE BAYES

Probability-based



NAÏVE BAYES

Assumes
independence

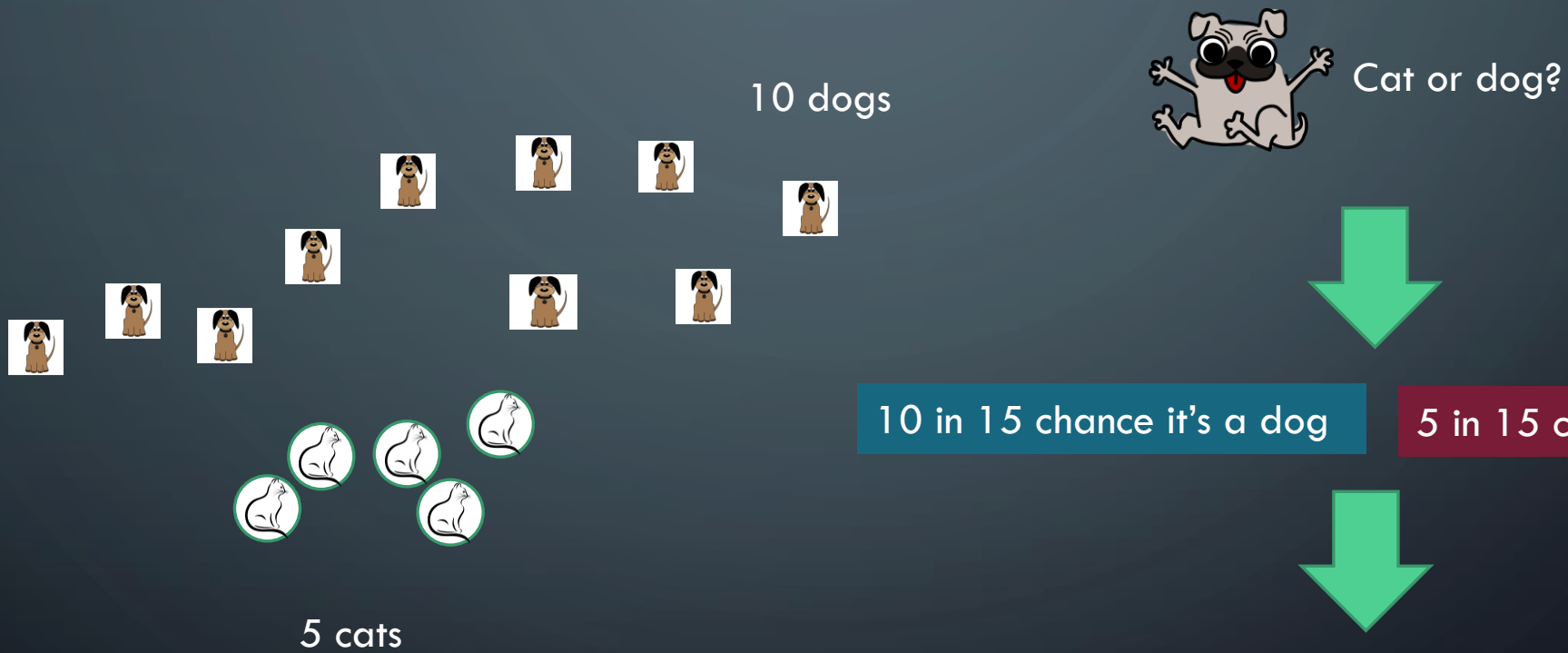


Needs
little data

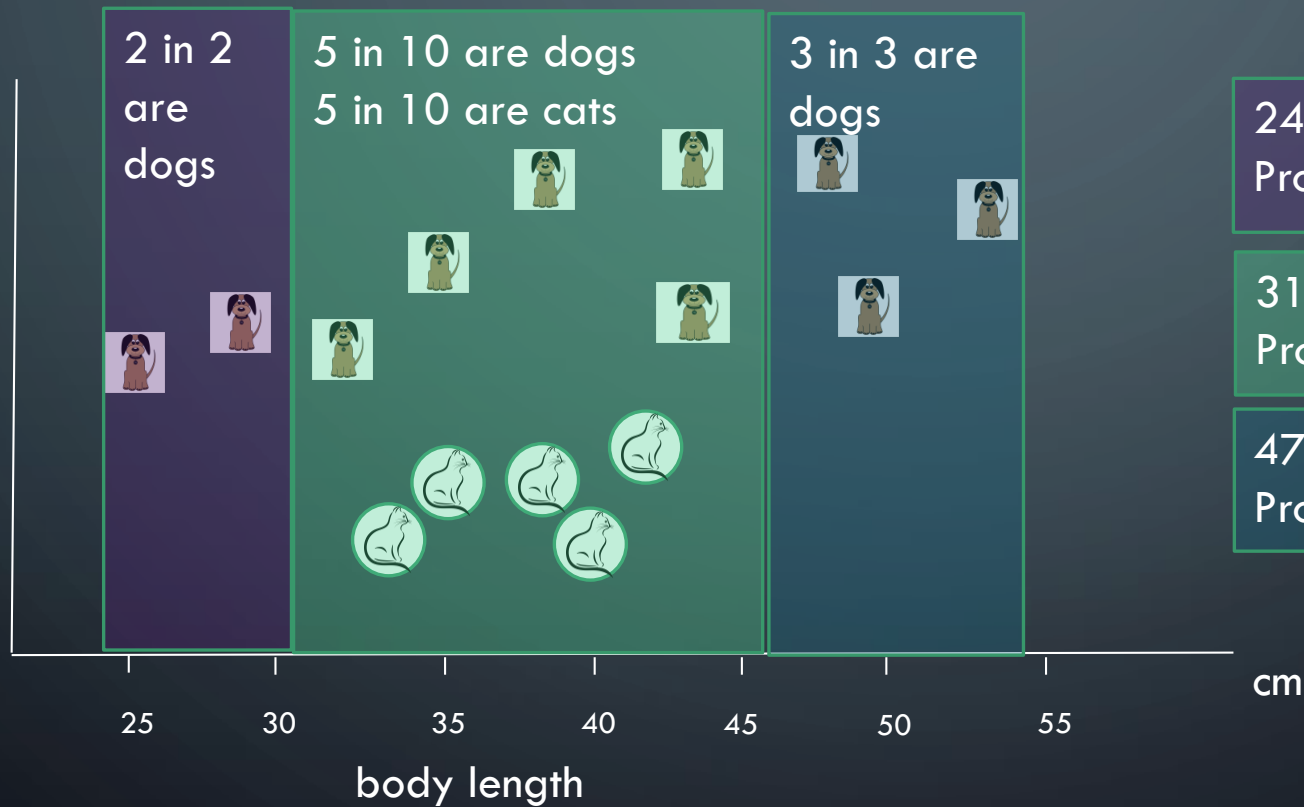
Simple &
fast

Insensitive to
irrelevant
features

PROBABILITY



PROBABILITY BY FEATURE

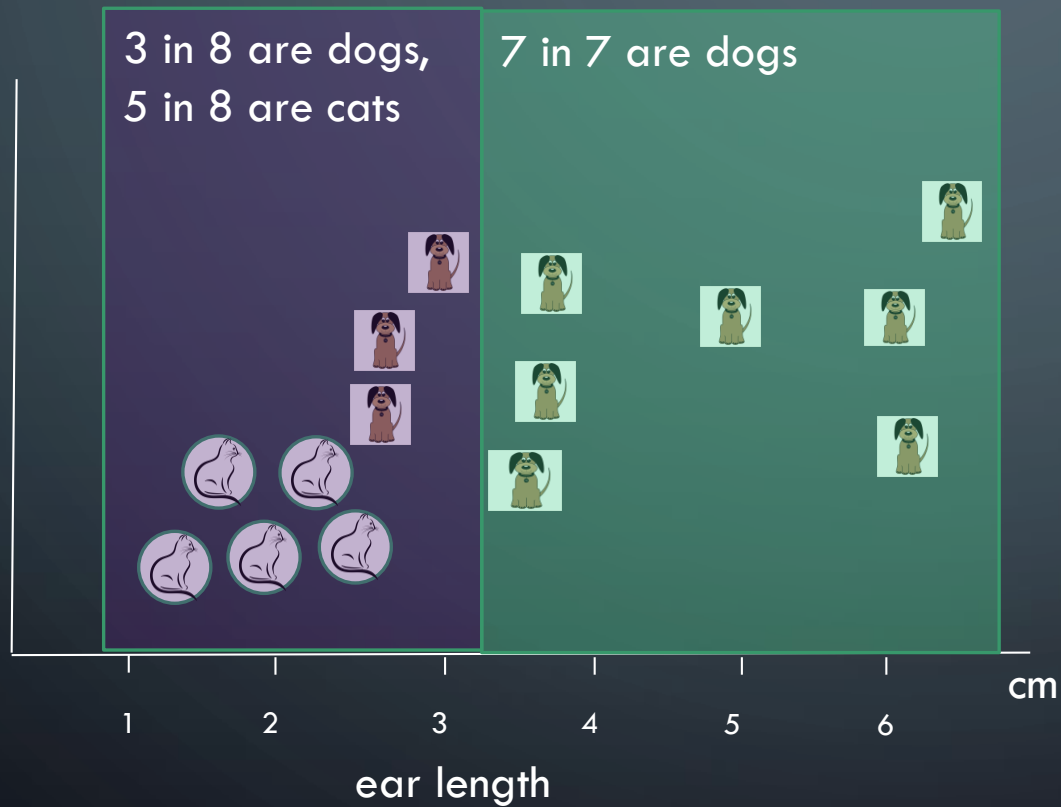


24 – 30cm:
Probability of dog 1, cat 0

31 – 46cm:
Probability of dog $\frac{1}{2}$, cat $\frac{1}{2}$

47 – 54cm:
Probability of dog 1, cat 0

PROBABILITY BY FEATURE



1 – 3cm:
Probability of dog $3/8$, cat $5/8$

4 – 7cm:
Probability of dog 1, cat 0

JOINT PROBABILITY

1 – 3cm:

Probability of dog $3/8$, cat $5/8$

4 – 7cm:

Probability of dog 1, cat 0

24 – 30cm:

Probability of dog 1, cat 0

31 – 46cm:

Probability of dog $1/2$, cat $1/2$

47 – 54cm:

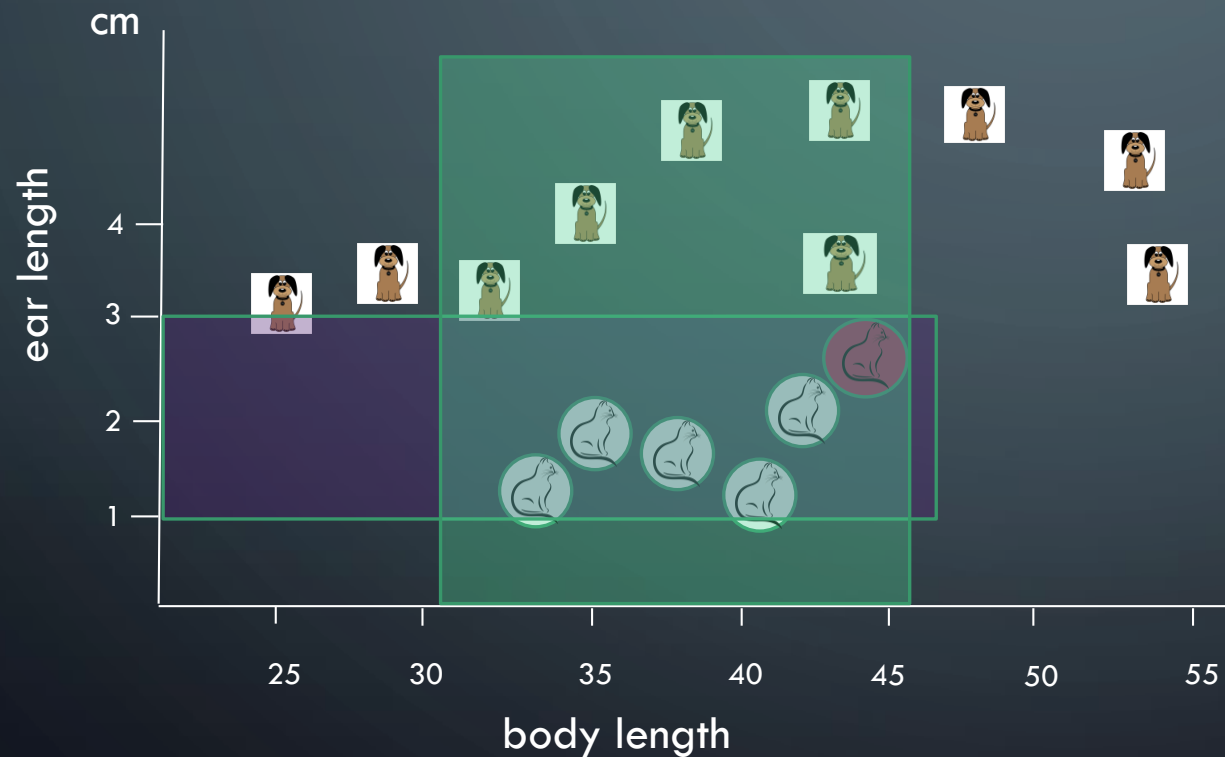
Probability of dog 1, cat 0

P(dog)	ears 1 – 3	ears 4 – 7
body 24-30	$3/8 * 1 = 0.375$	$1 * 1 = 1$
body 31-46	$3/8 * 1/2 = 0.188$	$1 * 1/2 = 0.5$
body 47-54	$3/8 * 1 = 0.375$	$1 * 1 = 1$

P(cat)	ears 1 – 3	ears 4 – 7
body 24-30	$5/8 * 0 = 0$	$0 * 0 = 0$
body 31-46	$5/8 * 1/2 = 0.313$	$0 * 1/2 = 0$
body 47-54	$5/8 * 0 = 0$	$0 * 0 = 0$

NAÏVE BAYES

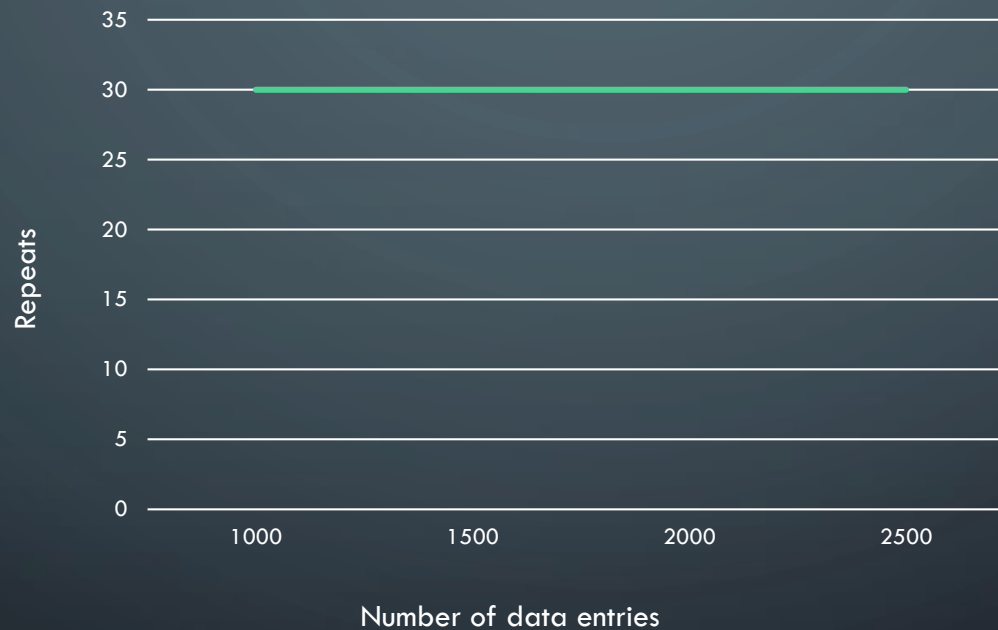
What are the probabilities the new case is a cat?



P(dog)	ears 1 – 3
body 31-46	0.188

P(cat)	ears 1 – 3
body 31-46	0.313

NAÏVE BAYES

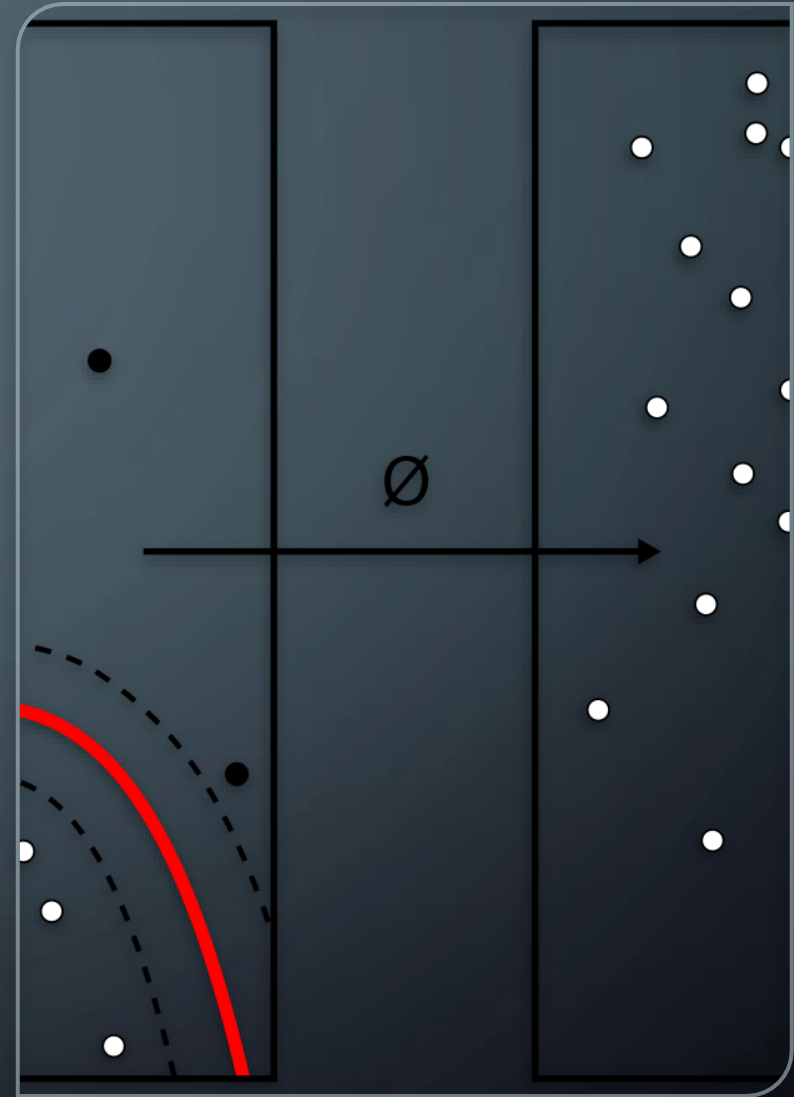


$$O(v * c)$$

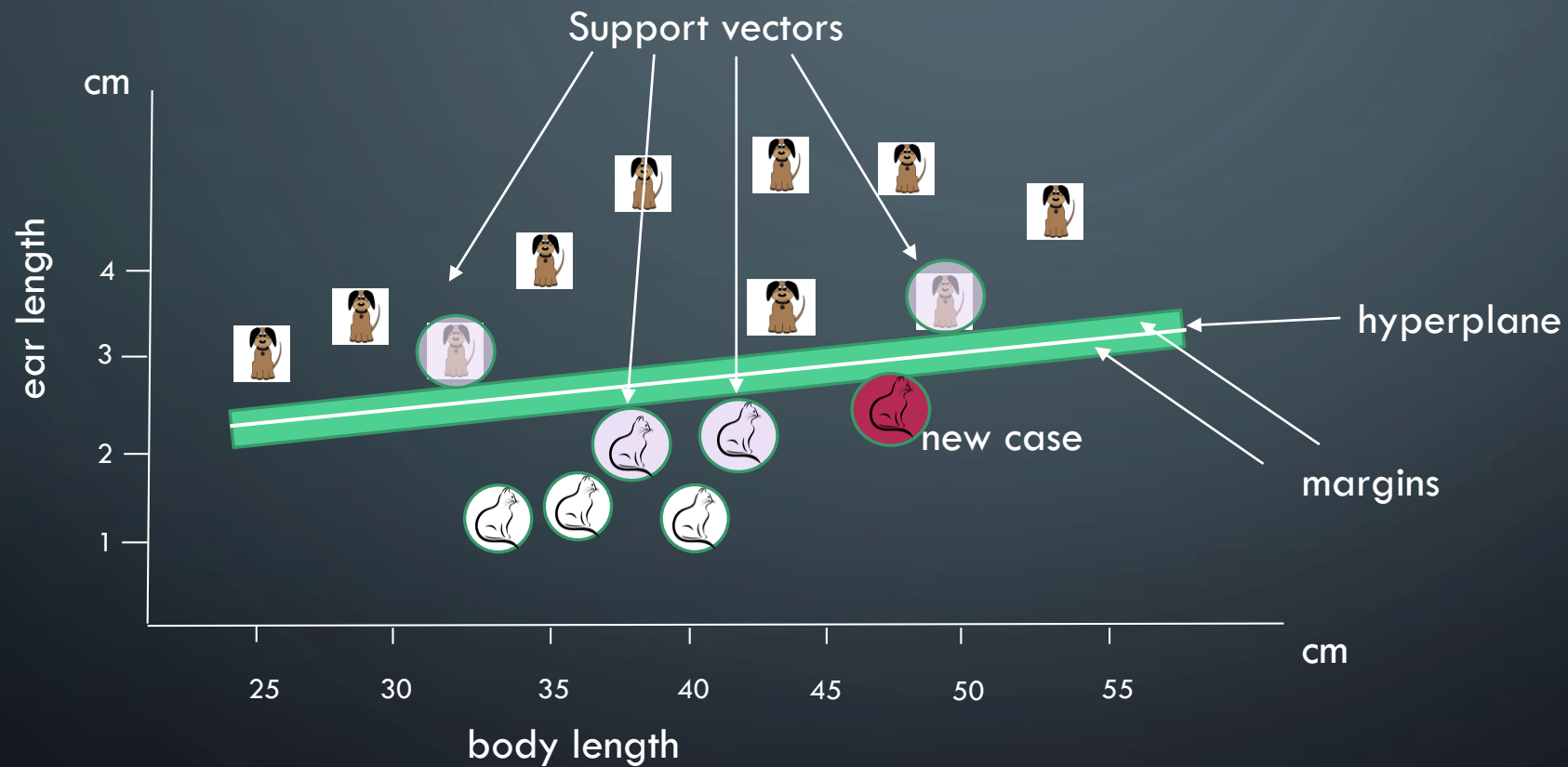
(v is number of variables,
assuming $v=3$
 c is the number of values,
assuming 10)

SUPPORT VECTOR MACHINES

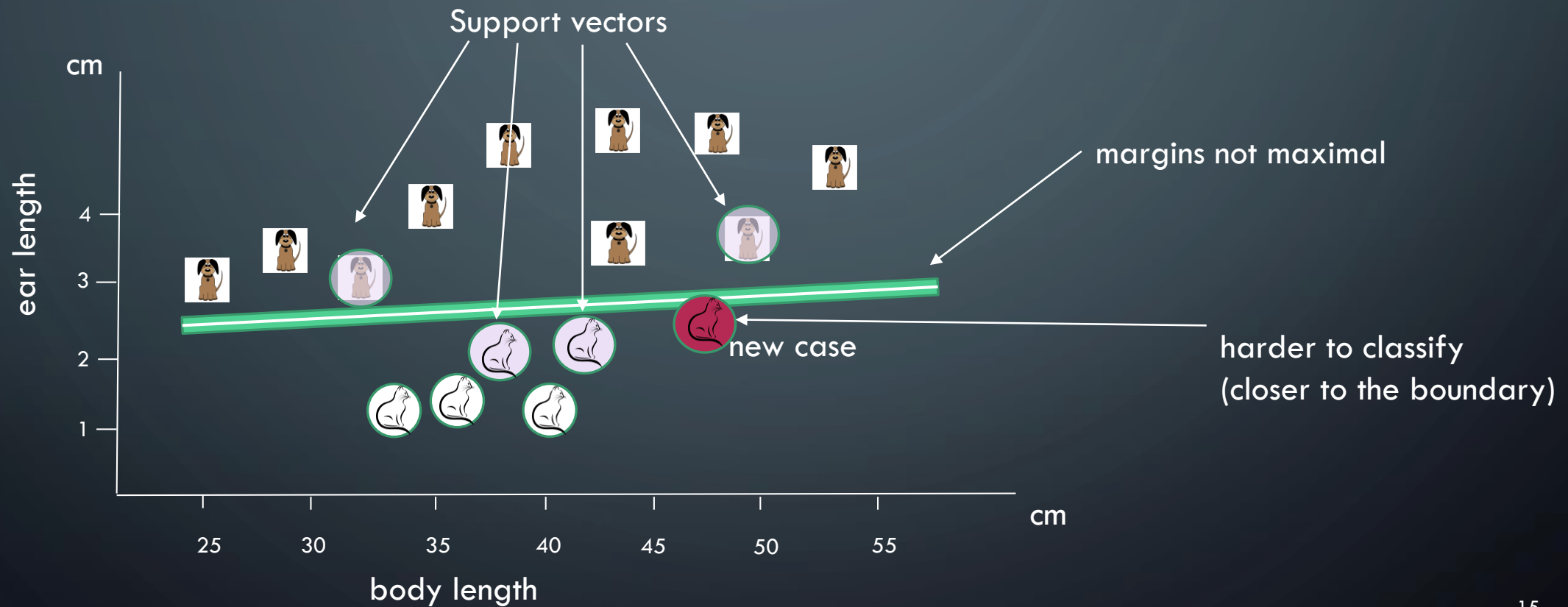
Finds the borders between groups



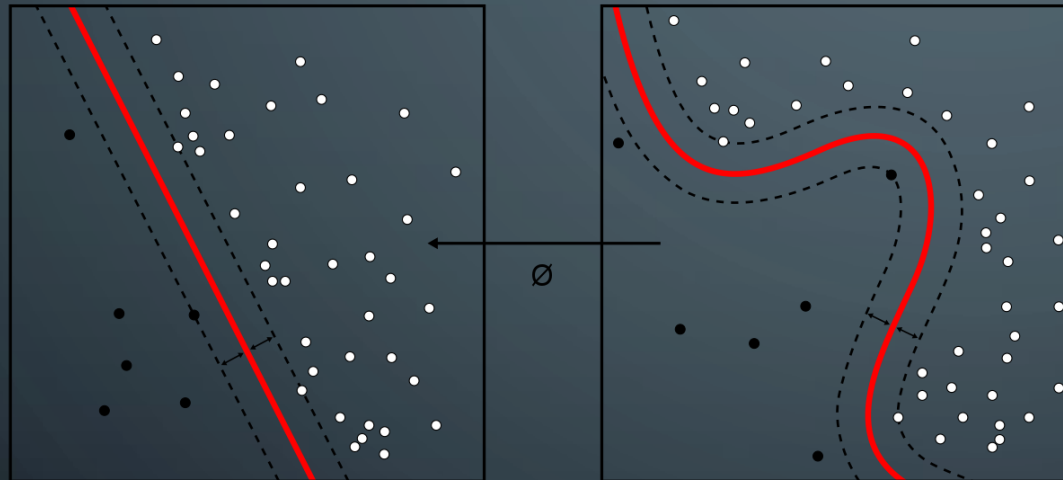
SUPPORT VECTOR MACHINES



SUPPORT VECTOR MACHINES



SUPPORT VECTOR MACHINES (SVM)



linear

?

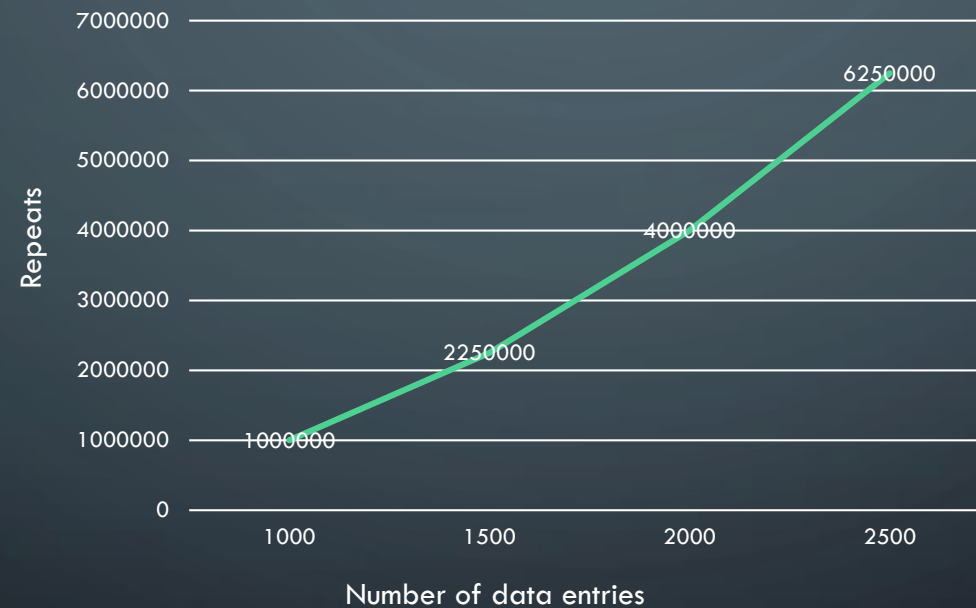
'Kernel' =
function

polynomial
(homogeneous)

polynomial
(inhomogeneous)

hyperbolic

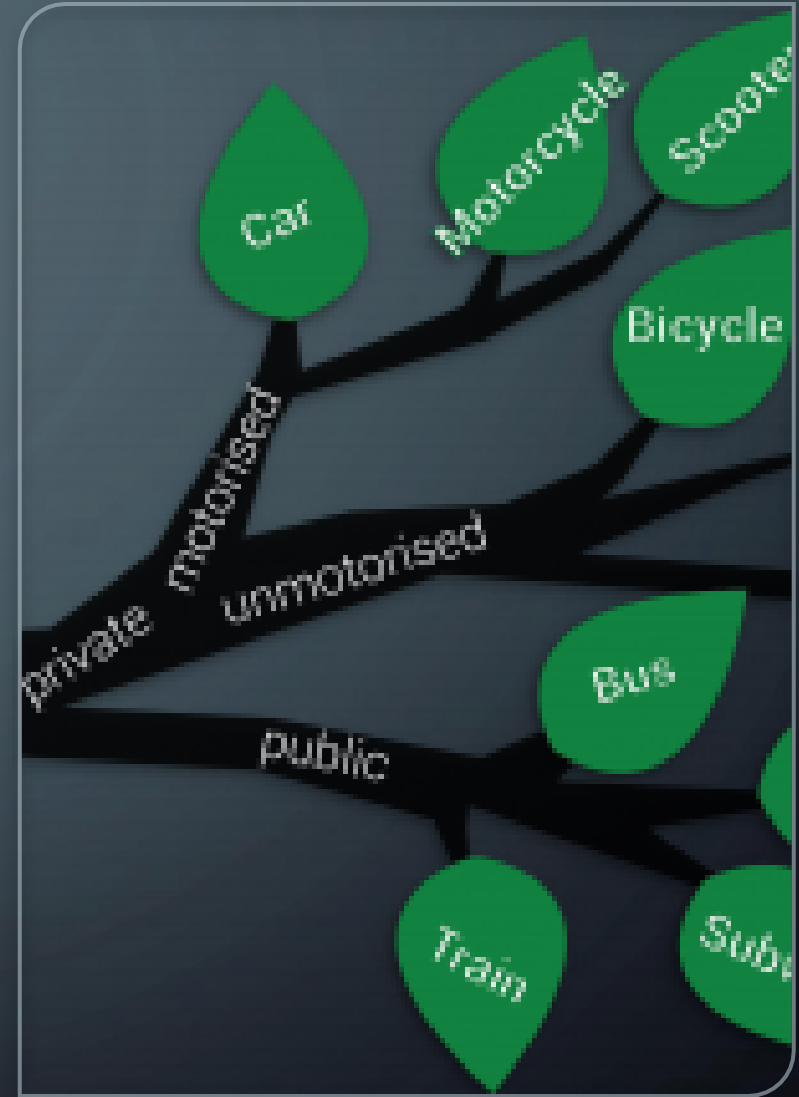
SVM - COMPLEXITY



$O(n^3)$

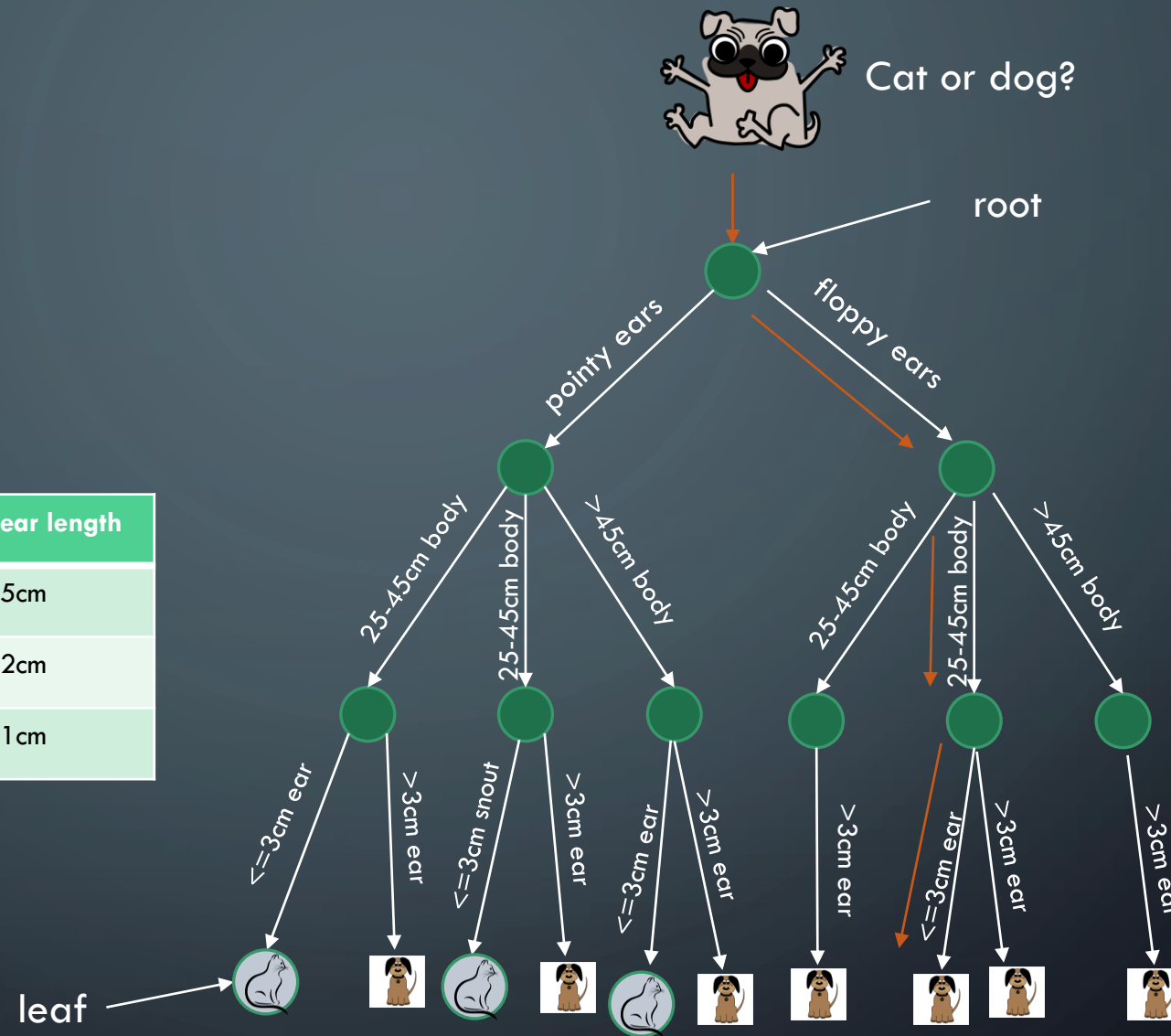
DECISION TREE

.. and random forest



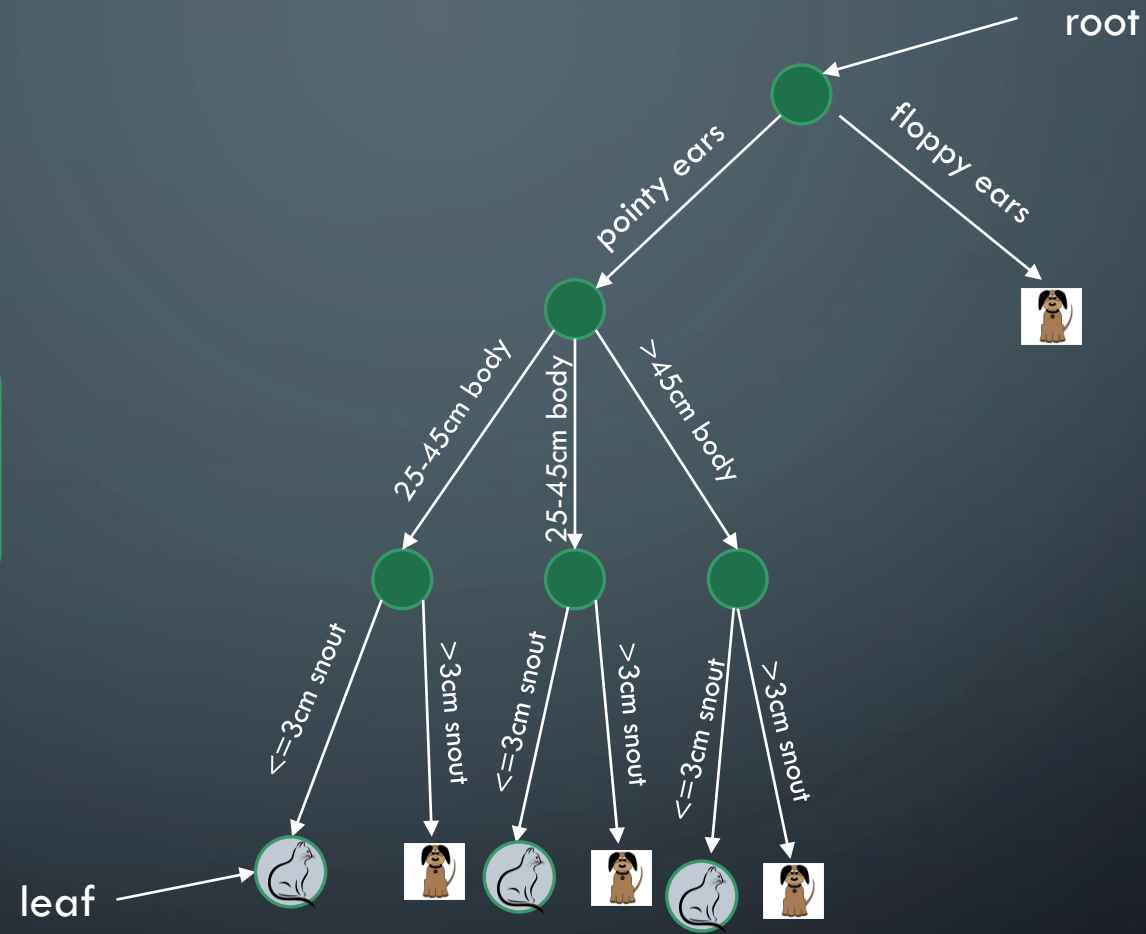
DECISION TREE

type	length	ears	ear length
dog	60cm	pointy	5cm
cat	45cm	pointy	2cm
dog	35cm	floppy	1cm

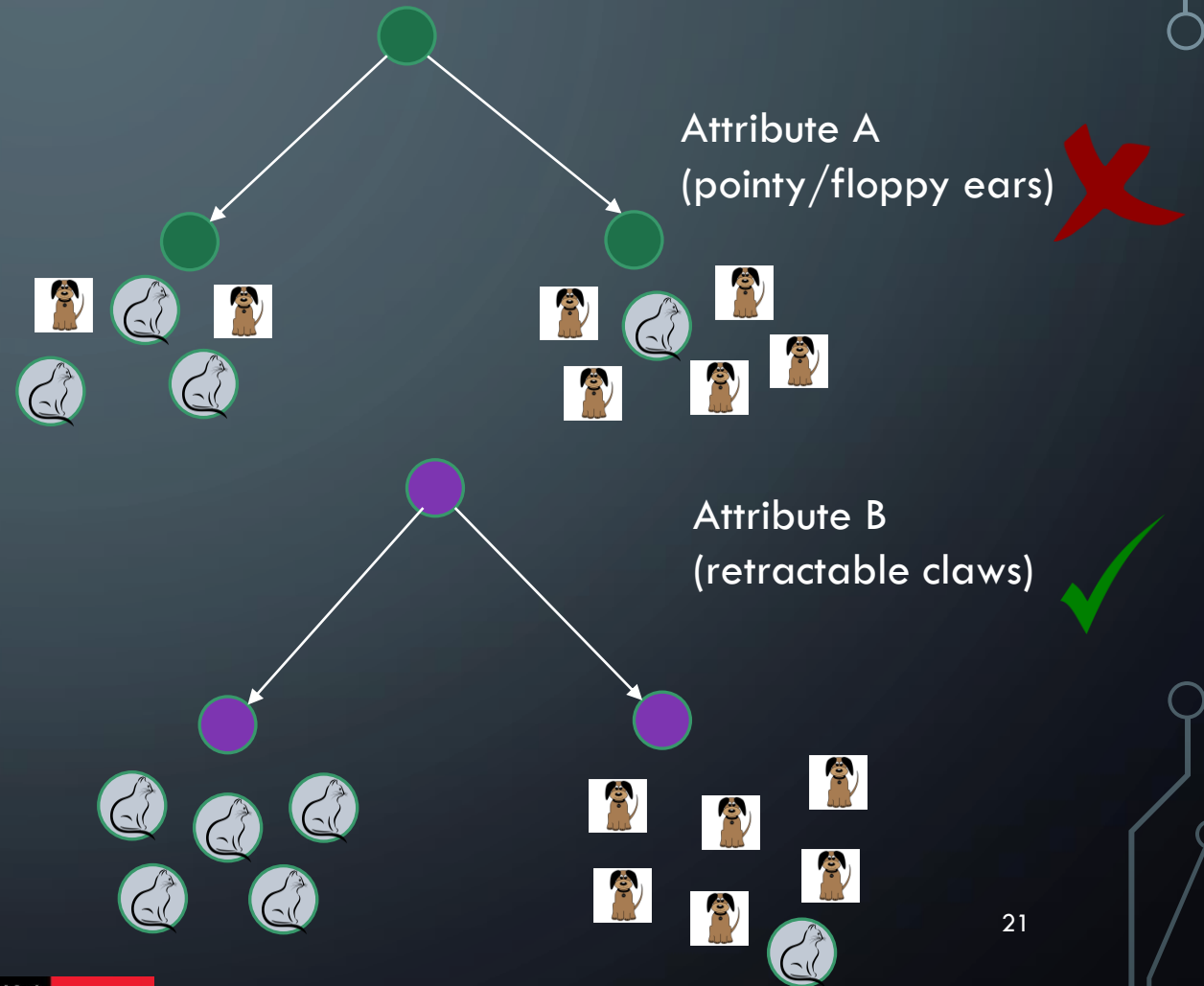
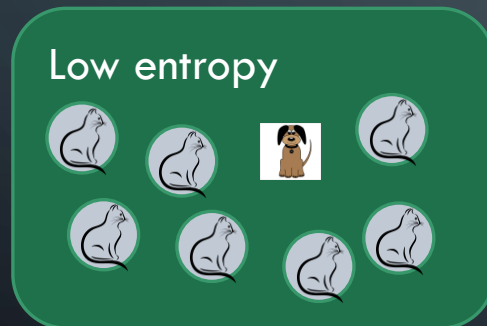
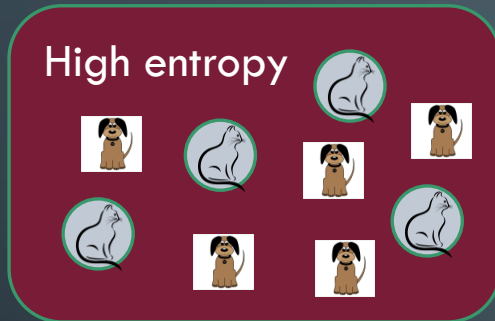


DECISION TREE

This simplified model contains the exact same information



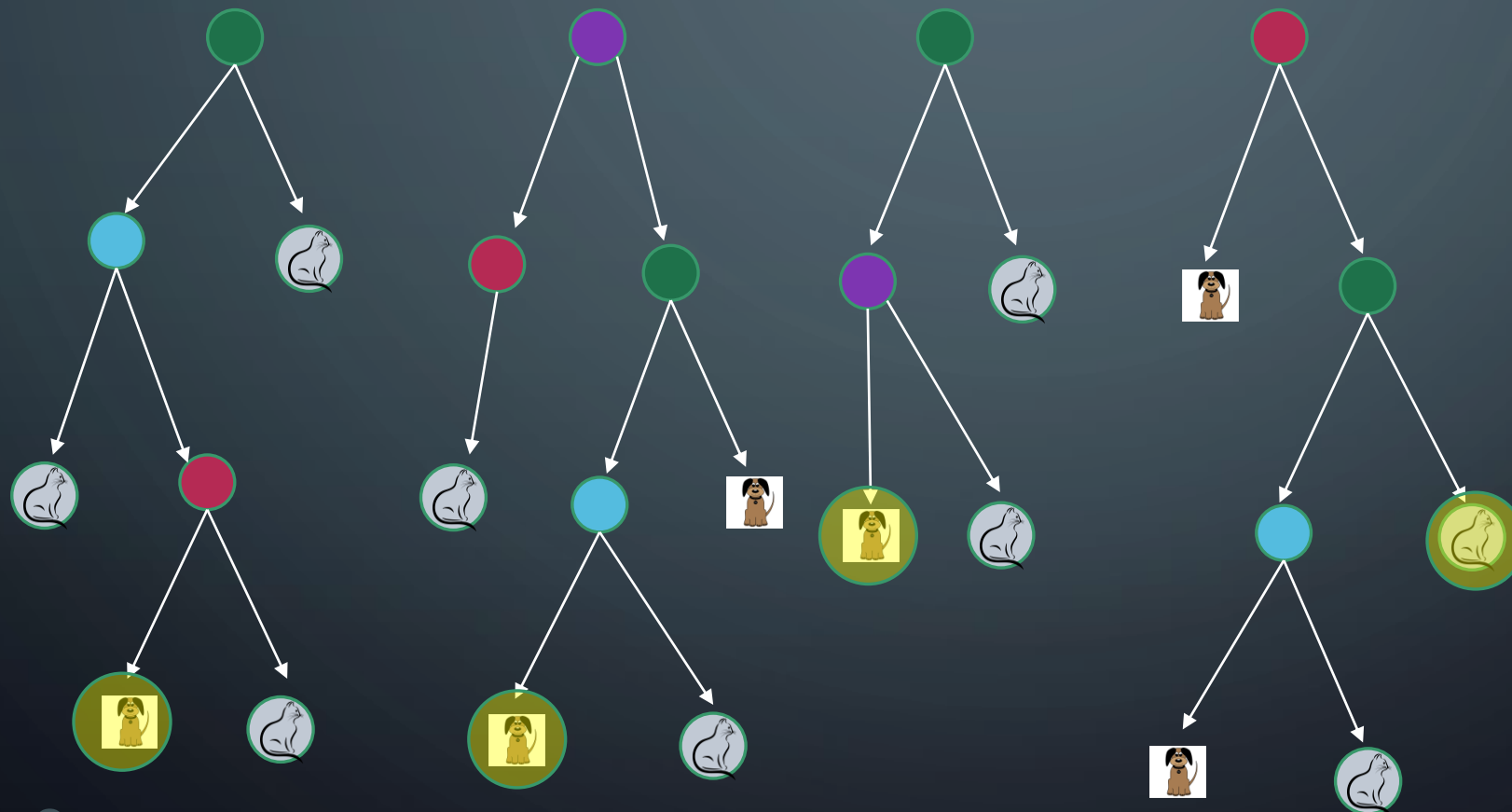
DECISION TREE



RANDOM FOREST

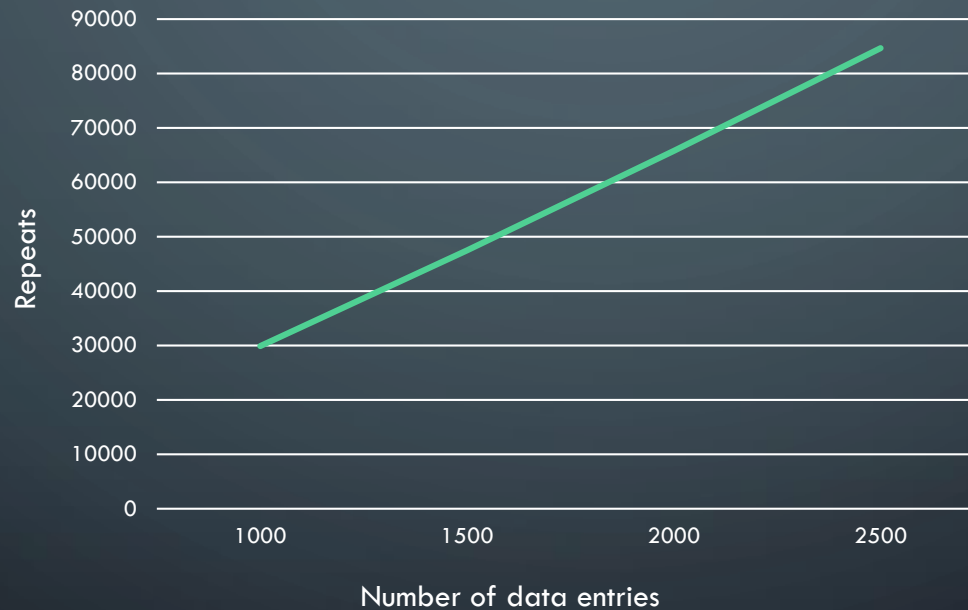


Cat or dog?



- Attribute A
(e.g. retractable claws)
- Attribute B
- Attribute C
- Attribute D

RANDOM FOREST COMPLEXITY



$$O(v * n \log(n))$$

(v is number of variables,
assuming $v=3$)

SUMMARY

- Classifiers decide how to **categorise** a sample based on its features.
- They build a model based on **features** in the data.
- The model can then be applied to classify **new data** we have never seen before.
- Prominent classifiers are **SVM, Naive Bayes, Decision Trees** and **Random Forest**.
 - Naive Bayes and Decision Trees are **fast**.
 - SVM and Random Forests are more **accurate** but need **more computation**.
 - SVM needs **parametrisation**: We have to choose a **kernel**.